

ALGORITHM TheLastAstronaut

INPUT: N (Jumlah astronot), K (Nilai K awal)

OUTPUT: Urutan astronot dieliminasi dan astronot terakhir bertahan

DECLARE astronauts AS ARRAY OF INTEGER OF SIZE N

DECLARE active_count AS INTEGER = N

DECLARE current_index AS INTEGER = 0

// Inisialisasi daftar astronot dari 1 hingga N

FOR i FROM 0 TO N-1 DO

 astronauts[i] = i + 1

END FOR

PRINT "Urutan Astronot yang Dieliminasi:"

WHILE active_count > 1 DO

 DECLARE steps AS INTEGER = 0

 // Hitung K astronot yang masih aktif

 WHILE steps < K - 1 DO

 IF astronauts[current_index] != 0 THEN

 steps = steps + 1

 END IF

 current_index = (current_index + 1) MOD N

 END WHILE

 // Pastikan posisi mendarat pada astronot yang aktif (non-nol)

 WHILE astronauts[current_index] == 0 DO

 current_index = (current_index + 1) MOD N

 END WHILE

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// Eliminasi astronot
DECLARE eliminated AS INTEGER = astronauts[current_index]
PRINT eliminated
astronauts[current_index] = 0 // Tandai astronot tereliminasi
active_count = active_count - 1

// Update nilai K berdasarkan paritas astronot yang dieliminasi
IF eliminated MOD 2 == 0 THEN
    K = K + 2
ELSE
    K = K - 1
END IF

// Pastikan nilai K tidak kurang dari 2
IF K < 2 THEN
    K = 2
END IF

IF active_count > 1 THEN
    REPEAT
        current_index = (current_index + 1) MOD N
    UNTIL astronauts[current_index] != 0
    END IF
END WHILE

// Cari astronot terakhir yang bertahan
FOR i FROM 0 TO N-1 DO
    IF astronauts[i] != 0 THEN
        PRINT "Astronot Terakhir yang Bertahan: ", astronauts[i]
    END IF

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END FOR

END ALGORITHM